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**Chapter 5**

2. Given the speeds listed in Fig. 5-1, is it possible to scan documents from a scanner and transmit them over an 802.11g network at full speed? Defend your answer.

5. A DMA controller has five channels. The controller is capable of requesting a 32-bit word every 40 nsec. A response takes equally long. How fast does the bus have to be to avoid being a bottleneck?

8. Suppose that a computer can read or write a memory word in 5 nsec. Also suppose that when an interrupt occurs, all 32 CPU registers, plus the program counter and PSW are pushed onto the stack. What is the maximum number of interrupts per second this machine can process?

12. A typical printed page of text contains 50 lines of 80 characters each. Imagine that a certain printer can print 6 pages per minute and that the time to write a character to the printer's output register is so short it can be ignored. Does it make sense to run this printer using interrupt-driven I/O if each character printed requires an interrupt that takes 50 μ sec all-in to service?

14. In which of the four I/O software layers is each of the following done.
    (a) Computing the track, sector, and head for a disk read.
    (b) Writing commands to the device registers.
    (c) Checking to see if the user is permitted to use the device.
    (d) Converting binary integers to ASCII for printing.

20. RAID level 3 is able to correct single-bit errors using only one parity drive. What is the point of RAID level 2? After all, it also can only correct one error and takes more drives to do so.

22. Compare RAID level 0 through 5 with respect to read performance, write performance, space overhead, and reliability.

28. Consider a magnetic disk consisting of 16 heads and 400 cylinders. This disk has four 100-cylinder zones with the cylinders in different zones containing 160, 200, 240, and 280 sectors, respectively. Assume that each sector contains 512 bytes, average seek time between adjacent cylinders is 1 msec, and the disk rotates at 7200 RPM. Calculate the (a) disk capacity, (b) optimal track skew, and (c) maximum data transfer rate.

31. Disk requests come in to the disk driver for cylinders 10, 22, 20, 2, 40, 6, and 38, in that order. A seek takes 6 msec per cylinder. How much seek time is needed for
    (a) First-come, first served.
    (b) Closest cylinder next.
    (c) Elevator algorithm (initially moving upward).